CRYPTOCard

CRYPTO-Server™ 6.x

3rd Party Integration

PHP, C, Java, Ruby, & Apache RADIUS Client(s)

Implementation Guide

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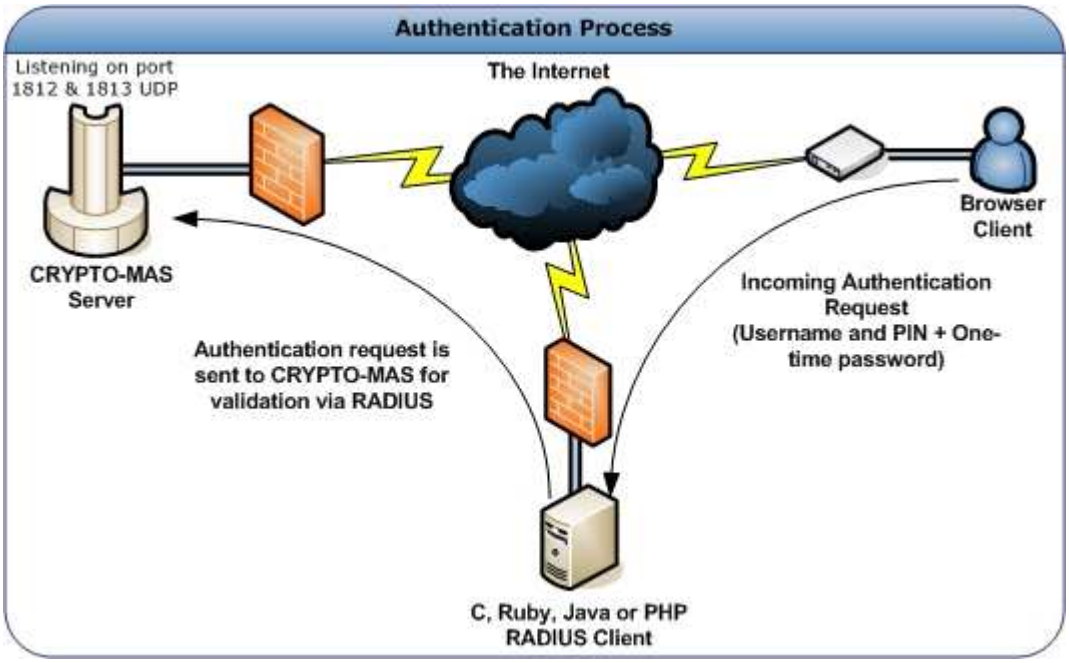
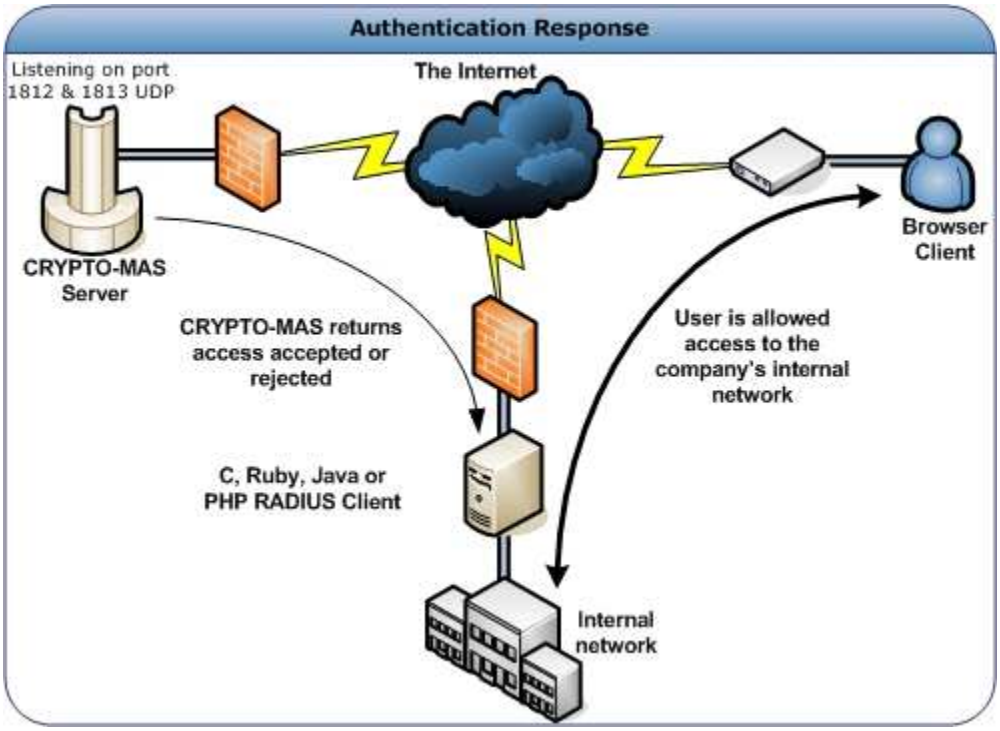
C, Ruby, Java & PHP RADIUS Client Overview

This document presents an overview and basic steps in configuring C or PHP RADIUS web based client(s) for use with CRYPTO-MAS and CRYPTOCard tokens.

The C, Ruby, Java or PHP RADIUS Client(s) is used to send RADIUS authentication requests to a CRYPTO-MAS server. CRYPTO- MAS works in conjunction with the C or PHP RADIUS Client(s) to replace static passwords with strong two-factor authentication that prevents the use of lost, stolen, shared, or easily guessed passwords when establishing a connection to gain access to protected resources.





Note: A Front page needs to be designed to display a username and One-time password (OTP) box for C (.cgi) and PHP (.php) for it to be a functional RADIUS authentication method for users.

PHP RADIUS Client

1. Testing environment: Windows Server 2003, IIS 6 with PHP 4 extensions from http://www.php.net/downloads.php, version PHP 4.4.4.

2. Download the PHP radius DLL (php_radius.dll) from http://pecl4win.php.net/ext.php/php_radius.dll. Here you are offered a choice of various versions for download, depending on the version of PHP that you are using.

3. This DLL should be placed in your PATH, and the file permissions should be adjusted to allow the PHP libraries to access it. That is, read and execute permissions are needed for the Internet Guest Account, usually IUSR_<MACHINE_NAME>.

4. A good example PHP script can be found at http://pecl.php.net/package/radius in the radius-1.2.4.tgz package. The file radius-auth.php in the examples directory is a good starting point for interacting with a MAS server over RADIUS.

5. The file radius-auth.php should be placed in your IIS web-hosting directory. The following changes are beneficial:

```
$username = 'john';
$password = '';
$radserver = '192.168.10.165';
$radport = 1812;
$sharedsecret = 'testing123';
$auth_type = 'pap';
//$auth_type = 'chap';
//$auth_type = 'mschapv1';
//$auth_type = 'mschapv2';
```

That is set the username, password, radserver, sharedsecret, and auth_type variables to match those expected by the CRYPTO-MAS. Also, in order to see the return data from the CRYPTO-MAS, you may want to add the following code:

```
default:
    $id = radius_cvt_string($data);
        echo "Data: [$id]<br>\n";
```

Add it to the bottom of the while loop's switch statement, in order to dump out the challenge messages from the CRYPTO-MAS.

C RADIUS Client

1. Testing environment: Red Hat Enterprise Linux 4.

   The following project http://freshmeat.net/projects/radiusclient/ contains a link where the radius client for C can be downloaded. Choose the following version: ftp://ftp.cityline.net/pub/radiusclient/radiusclient-0.3.2.tar.gz.

2. Uncompress and un-tar the radiusclient-0.3.2.tar.gz file to a directory of your choosing. From this directory, the C library can be configured, built and installed by issuing:

```
# ./configure
# make
# make install
```

3. In src/radexample.c is a small example of how to use the library lib/libradclient.a.

You only need to include radiusclient.h into your applications and link against the library. For this example to work your radiusclient config file has to be setup properly. This involves the following:

4. In the etc directory, type "make install". This will install a series of files to /usr/local/etc/radiusclient/.

5. Edit the file /usr/local/etc/radiusclient/servers to add the hostname/IP of your radius server, and its shared secret. For example:

```
#Server Name or Client/Server pair                Key
#----------------                                 ---------------
#portmaster.elemental.net                         hardlyasecret
#portmaster2.elemental.net                        donttellanyone
192.168.10.165                                    testing123
```

6. Edit the following file: /usr/local/etc/radiusclient/radiusclient.conf

   Change the authserver entry to your Radius ip and auth port number.
   For example: authserver 192.168.10.165:1812
   Change the acctserver entry to your Radius ip and accounting port number.
   For example: acctserver 192.168.10.165:1813

7. The test script that links against the library lib/libradclient.a can be run as follows, from the src directory:

```
[root@james-redhat src]# ./radexample
login: john
Password:
"john" RADIUS Authentication OK
```

Notes: Check /var/log/messages for possible errors messages that may be the result of an incorrectly configured library**.**

Java RADIUS Client

1. Testing environment – Red Hat Enterprise Linux 4.
2. Download jradius-client-2.0.0.tar from: http://jradius-client.sourceforge.net/. It will contain a JAR that can perform the RADIUS authentications (jradius-client.jar).
3. This jar also requires that the java-getopt-1.0.13.jar be in the class path as well. This jar is free to download as well.
4. The java client can be run from the command line as follows:

```
[root@james-redhat jradiusclient]# java -classpath jradius-
client.jar:java-getopt-1.0.13.jar
net.sourceforge.jradiusclient.TestRadiusClient -s 192.168.10.165 -S
testing123
```

   Where the value following the -s is the IP/hostname of the RADIUS server, and the value following the -S is the shared secret.

5. This will launch an interactive login session where the login credentials can be entered, for example:

```
[root@james-redhat jradiusclient]# java -classpath jradius-
client.jar:java-getopt-1.0.13.jar
net.sourceforge.jradiusclient.TestRadiusClient -s 192.168.10.165 -S
testing123
Performing tests using basic classes:
Username: john
Password: 613-6977
Authentication method [PAP | chap]: PAP
Additional Attributes? [y|N]:n
TestRadiusClient: User john authenticated
Response Packet Attributes
        Type    Value
TestRadiusClient: User john got ACCOUNTING_RESPONSE response
Response Packet Attributes
        Type    Value
Another Basic Test [ Y | n ]: n
```

Note: This library tests authentication and accounting responses. It also supports challenge response as seen below:

```
[root@james-redhat jradiusclient]# java -classpath jradius-
client.jar:java-getopt-1.0.13.jar
net.sourceforge.jradiusclient.TestRadiusClient -s 192.168.10.165 -S
testing123
Performing tests using basic classes:
Username: john
Password: 123780-8534
Authentication method [PAP | chap]: PAP
Additional Attributes? [y|N]:N
TestRadiusClient: User john Challenged with Your PIN has expired. Your
new PIN is 915.
Re-authenticate using your new PIN and next PASSCODE.
Another Basic Test [ Y | n ]: Y
Username: john
Password: 915417-6731
Authentication method [PAP | chap]: PAP
Additional Attributes? [y|N]:N
TestRadiusClient: User john authenticated
Response Packet Attributes
        Type    Value
TestRadiusClient: User john got ACCOUNTING_RESPONSE response
Response Packet Attributes
        Type    Value
Another Basic Test [ Y | n ]: n
```

Ruby RADIUS Client

1. Testing environment: Red Hat Enterprise Linux 4, Ruby version 1.8.5.

   It is assumed ruby is installed on your system. If it is not, you may download Ruby from http://www.ruby-lang.org/en/.

2. The radius library for Ruby can be downloaded from http://www.nodewarrior.org/radiuslib/. Download the RADIUSlib-0.5.tar.gz file and uncompress and un-tar it to a directory of your choosing.

3. From here configuring and installing the ruby library can be done in three steps from the root directory of the project:

```
# ruby install.rb config
# ruby install.rb setup
# ruby install.rb install
```

4. The library is now setup in the following location: /usr/local/lib/ruby/site_ruby/1.6/radius

5. Authenticating using the Ruby radius library can be accomplished by running the radtest.rb Ruby script found in the sample directory. The radtest program works with the following syntax:

```
Syntax: radtest.rb [-cpov] [-d <dict>] [-f <filename>] [<username>]
[<password>] <server[:port]> <secret>
  Options:

  -c, --chap            Use CHAP to authenticate.
  -p, --pap             Use PAP to authenticate (default).
  -o, --accounting      Send an accounting request, not authentication.
  -f, --file            Read attributes from <filename>. A '-' can be
                        supplied to read attributes from STDIN.
  -d, --dict            The name of a dictionary (default:
/etc/raddb/dictionary)
  -v, --verbose         Increase verbosity.
  -h, --help            This message.

  Example: radtest.rb -c testuser testpass radiusserver MySeCrEt
```

6. Additional example using a CRYPTOServer:

```
[root@james-redhat sample]# ruby radtest.rb -p -d
/usr/local/etc/radiusclient/dictionary john 307-
8319 192.168.10.165:1812 testing123
Sending request to 192.168.10.165:1812....
Reply received from 192.168.10.165, port 1812
Successful Authentication
```

Apache Radius Client

Apache 1.3 on Red Hat Linux Enterprise 4 was tested using its mod_auth_radius module. The source file for this module (mod_auth_radius.c) can be obtained from http://www.freeradius.org/mod_auth_radius/. To install Apache with mod_auth_radius go to the Apache root directory, and type:

./configure --add-module=LOCATION/mod_auth_radius.c

Where LOCATION is the directory where the module source file resides. Then, type:

make

Then type:

make install

This will create a radius aware Apache server. In order to protect a directory with Radius authentication, some extra information must be added into the httpd.conf file. For example, to protect a directory called /secure add the following to the bottom of httpd.conf:

```
#####################################################################
#
# Add to the BOTTOM of httpd.conf
# If we're using mod_auth_radius, then add it's specific
# configuration options.
#
<IfModule mod_auth_radius.c>

#
# AddRadiusAuth server[:port] <shared-secret> [ timeout [ : retries ]]
#

# Use localhost, the old RADIUS port, secret 'testing123',
# time out after 5 seconds, and retry 3 times.
#AddRadiusAuth localhost:1645 testing123 5:3
AddRadiusAuth 192.168.10.165:1812 testing123 5:3

#
# AuthRadiusBindAddress <hostname/ip-address>
#
# Bind client (local) socket to this local IP address.
# The server will then see RADIUS client requests will come from
# the given IP address.
#
# By default, the module does not bind to any particular address,
# and the operating system chooses the address to use.
#

#
# AddRadiusCookieValid <minutes-for-which-cookie-is-valid>
#
# the special value of 0 (zero) means the cookie is valid forever.
```

```
#
AddRadiusCookieValid 5
</IfModule>

###################################################################
#
#  A sample per-directory access-control configuration.  This may
#  go into httpd.conf.  When used in an '.htaccess' file, the
#  text BETWEEN the <Location /secure/> </Location> directives
#  should go into the .htaccess file.
#
#  If you want to be sure it only applies when mod_auth_radius is used,
#  you can also wrap it in an <IfModule> directive, as above.
#
<Location /secure/>

#
# Use basic password authentication.
# AuthType Digest won't work with RADIUS authentication.
#
AuthType Basic

#
# Tell the user the realm to which they're authenticating.
# This string should be configured for your site.
#
AuthName "RADIUS authentication"

#
# don't use 'mod_auth'.
# You might want to disable other authentication types here.
# You can get a similar effect by commenting out the
# 'AddModule mod_auth_*' lines, previously in httpd.conf
#
AuthAuthoritative off

#
# Use mod_auth_radius for all authentication, and make the responses
# from it authoritative.
#
AuthRadiusAuthoritative on

#
# Make a local variation of AddRadiusCookieValid.  The server will
choose
# the MINIMUM of the two values.
#
# AuthRadiusCookieValid <minutes-for-which-cookie-is-valid>
#
AuthRadiusCookieValid 5
```

```
#
# Set the use of RADIUS authentication at this <Location>"
#
# Globally set the RADIUS authentication active.
#
#
# If there is a directory which you do NOT want to have RADIUS
# authentication for, then use a <Directory> or <Location> directive,
# and set "AuthRadiusActive Off".  The default is "On".
#
AuthRadiusActive On

#
# require that mod_auth_radius return a valid user, otherwise
# access is denied.
#
require valid-user

#
# end of the per-location directives
#
</Location>
```

Note that the "AddRadiusAuth 192.168.10.165:1812 testing123 5:3" directive should be changed to reflect the IP, port and shared secret of your own radius server. Also the directive "Location /secure/>" should be changed to the name of the directory on your website that you are trying to protect.

Please see http://www.freeradius.org/mod_auth_radius/README for other helpful hints.

Solution Overview

Summary	
Product Name(s)	PHP RADIUS Client C RADIUS Client Java RADIUS Client Ruby RADIUS Client Apache RADIUS
Vendor Site(s)	http://pecl4win.php.net/ext.php/php_radius.dll ftp://ftp.cityline.net/pub/radiusclient/radiusclient-0.3.2.tar.gz http://jradius-client.sourceforge.net/ http://www.nodewarrior.org/radiuslib/ http://www.freeradius.org/mod_auth_radius/
Supported SSH Client Software	Linux command line interface Internet Explorer 6 or higher Mozilla Firefox 1.5 or higher
Authentication Method	RADIUS Authentication
Supported RADIUS Functionality for PHP RADIUS Client Browser Connection	
RADIUS Authentication Encryption	PAP MSCHAPv2
Authentication Method	One-time password Challenge-response Static Password
New PIN Mode	User-changeable Alphanumeric 4-8 digit PIN User-changeable Numeric 4-8 digit PIN Server-changeable Alphanumeric 4-8 digit PIN Server-changeable Numeric 4-8 digit PIN
Supported RADIUS Functionality for C RADIUS Client Browser Connection	
RADIUS Authentication Encryption	PAP MSCHAPv2
Authentication Method	One-time password Static Password
New PIN Mode	User-changeable Alphanumeric 4-8 digit PIN User-changeable Numeric 4-8 digit PIN Server-changeable Alphanumeric 4-8 digit PIN Server-changeable Numeric 4-8 digit PIN

Supported RADIUS Functionality for Java RADIUS Client Browser Connection	
RADIUS Authentication Encryption	PAP MSCHAPv2
Authentication Method	One-time password Challenge-response Static Password
New PIN Mode	User-changeable Alphanumeric 4-8 digit PIN User-changeable Numeric 4-8 digit PIN Server-changeable Alphanumeric 4-8 digit PIN Server-changeable Numeric 4-8 digit PIN
Supported RADIUS Functionality for Ruby RADIUS Client Browser Connection	
RADIUS Authentication Encryption	PAP MSCHAPv2
Authentication Method	One-time password Static Password
New PIN Mode	User-changeable Alphanumeric 4-8 digit PIN User-changeable Numeric 4-8 digit PIN Server-changeable Alphanumeric 4-8 digit PIN Server-changeable Numeric 4-8 digit PIN
Supported RADIUS Functionality for Apache RADIUS Client Browser Connection	
RADIUS Authentication Encryption	PAP MSCHAPv2
Authentication Method	One-time password Challenge-response Static Password
New PIN Mode	User-changeable Alphanumeric 4-8 digit PIN User-changeable Numeric 4-8 digit PIN Server-changeable Alphanumeric 4-8 digit PIN Server-changeable Numeric 4-8 digit PIN

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